

	x_i	y_i	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$	$y = Ax + B$
1	1	4.5	20.25	108.16	46.8	
	2	11	12.25	15.21	13.65	
	3	8	6.25	47.61	17.25	
	4	13.9	2.25	1	1.5	
	5	12	.25	8.41	1.45	
	6	14	.25	.81	-.45	
	7	18	2.25	9.61	4.65	
	8	25	6.25	102.01	25.85	
	9	19	12.25	16.81	14.35	
	10	23.6	20.25	75.69	39.15	
Sum	$(\sum x) 55$	$(\sum y) 149.9$	8.25	38.532	16.36	
			$(\sum x^2)$	$(\sum y^2)$	$\sum xy$	

$$\frac{SS_{xx} - SS_{yy}}{-30.282} < 0$$

$$B = \frac{(SS_{xx} - SS_{yy})}{2SS_{xy}} = -9.2549$$

$$a_1 = -B \pm B \sqrt{1 + \frac{1}{B^2}}$$

$$= (-(-9.2549) - (-9.2549) \sqrt{1 + \frac{1}{(-9.2549)^2}})$$

$$= 2.288$$

$$b_1 = \bar{y} - a_1 \bar{x} = 14.9 - 2.28(5.5)$$

$$= 2.36 \quad \checkmark$$

$$y = 2.288x + 2.36$$

Quality of fit:

$$r_1: \frac{1}{2} \{ (SS_{xx} + SS_{yy}) + \sqrt{(SS_{xx} - SS_{yy})^2 + 4(SS_{xy})^2} \} = 45.6823$$

$$r_2: \frac{1}{2} \{ (SS_{xx} + SS_{yy}) - \sqrt{(SS_{xx} - SS_{yy})^2 + 4(SS_{xy})^2} \} = 1.09976$$

$$m^2: \left(\frac{a_1 - a_2}{1 + a_1 a_2} \right)^2 = 0 \quad Q^2 = \left(\frac{r_1 - r_2}{r_1 + r_2} \right)^2 = 0.9082$$

$$a_v = \frac{SS_{xy}}{SS_{xx}} = \frac{16.36}{8.25} = 1.98 \quad b_v = \bar{y} - a_v \bar{x} = 14.9 - 5.5(1.98) = 4.01$$

$$y = 1.98x + 4.01 \quad \checkmark$$

$$m^2 = \left(\frac{1.98 - 2.288}{1 + 1.98(2.288)} \right)^2 = 0.0031$$

$$Q^2 = \frac{45.6823 - \sqrt{45.6823^2 \cdot (0.0031) + (1.09976)^2}}{45.6823 \sqrt{0.0031} + \sqrt{45.6823^2 \cdot (0.0031) + (1.09976)^2}} = \frac{42.983}{48.523} = (0.8858)^2 = 0.7846$$

$$c. \quad a_h = \frac{SS_{yy}}{SS_{xy}} = \frac{38.532}{16.36} = 2.355$$

$$b_h = \bar{y} - a_h \bar{x} = 14.9 - 5.5(2.355) = 1.9475$$

$$y = 2.355x + 1.9475 \quad \checkmark$$

$$m^2 = \left(\frac{2.355 - 2.288}{1 + 2.355(2.288)} \right)^2 = 1.1 \cdot 10^{-4}$$

$$Q^2 = \frac{45.6823 - \sqrt{45.6823^2 \cdot (1.1 \cdot 10^{-4}) + (1.09976)^2}}{45.6823 \sqrt{1.1 \cdot 10^{-4}} + \sqrt{45.6823^2 \cdot (1.1 \cdot 10^{-4}) + (1.09976)^2}} = \frac{44.58253}{46.78} = (0.9528)^2$$

$$= 0.908$$

2. $r = \frac{A\bar{z}}{(B+\bar{z})}$ 1) $\frac{1}{r} = \frac{B+\bar{z}}{A\bar{z}} \rightarrow \boxed{\frac{1}{r} = \frac{B}{A\bar{z}} + \frac{1}{A}}$

2) $r(B+\bar{z}) = A\bar{z}$
 $rB + r\bar{z} = A\bar{z}$
 $(r\bar{z} = A\bar{z} - rB) \div \bar{z}$
 $\boxed{r = A - \frac{rB}{\bar{z}}}$

1) $\frac{1}{\bar{z}}$	$\frac{1}{r}$	$(X-S_x)^2$	$(Y-S_y)^2$	$(X-S_x)(Y-S_y)$
.4	.65	.0870	.0016	0.01
.25	.0233	.0210	.0000585	0.00106
.1	.0154	.0000263	$2.12 \cdot 10^{-7}$	$2.86 \cdot 10^{-6}$
.05	.00913	.00304	$4.24 \cdot 10^{-5}$	.000359
.025	.00893	.00642	$4.92 \cdot 10^{-5}$	.000562
.01	.00781	.00905	.0000661	.000773
.005	.00667	.0100	.000040	.000929
.001	.00666	.0108	.0000477	.00103
.105	.0154	0.0189	.000194	0.00184
(S_x)	(S_y)	(SS_{xx})	(SS_{yy})	(SS_{xy})

$\lambda_1: (0.0189 + 0.000194) + \sqrt{(0.0189 - 0.000194)^2 + 4(0.00184)}$

$\boxed{= 0.0186}$

$\lambda_2: (0.0189 + 0.000194) - \sqrt{(0.0189 - 0.000194)^2 + 4(0.00184)}$

$\boxed{= 0.0000954}$

$Q^2 = \left(\frac{\lambda_1 \sqrt{n^2+1} - \sqrt{\lambda_1^2 n^2 + \lambda_2^2}}{\lambda_1 \sqrt{n^2+1} + \sqrt{\lambda_1^2 n^2 + \lambda_2^2}} \right)^2 = \boxed{0.998}$

$m^2 = 0$ because 1

$B = \frac{SS_{xx} - SS_{yy}}{2SS_{xy}} = \boxed{4.990}$ + because $SS_{xx} - SS_{yy} > 0$

$a_{\perp} = -B \sqrt{1 + \frac{1}{B^2}} = \boxed{0.1}$

2) $\bar{z} = \frac{S}{\bar{z}}$	r	$(X-S_x)^2$	$(Y-S_y)^2$	$(X-S_x)(Y-S_y)$
8	20	12.7	6181.891	-279.659
10.75	43	39.8	3094.141	-350.82
6.5	65	4.23	1130.641	-69.162
5.3	106	0.734	54.391	.6.319
2.8	112	2.7	178.891	-21.977
1.28	128	10	862.891	-92.917
.75	150	13.6	2639.391	-189.734
.165	165	18.3	4405.441	-283.961
7.443	98.625	12.755	2318.484	-160.239
(S_x)	(S_y)	(SS_{xx})	(SS_{yy})	(SS_{xy})

$\lambda_1 = 2329.567$

$\lambda_2 = 1.672$

$m^2 = 0$ because 1

$Q^2 = \boxed{0.997}$

$B_1 = \boxed{7.195}$

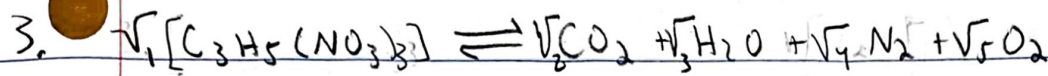
$a_{\perp} = -7.195 - \sqrt{1 + \frac{1}{(7.195)^2}}$
 $= \boxed{-14.459}$

Need provide A & B for both questions 1 & 2.

b). See back of page 2

b7 -6.
Need use non linear solver.

76/84



See back for matrix

$$C \rightarrow 3\nu_1 = \nu_2$$

$$\nu_1 = 1 \text{ so } \dots \boxed{\nu_2 = 3}$$

$$H \rightarrow 5\nu_1 = 2\nu_3$$

$$\boxed{\nu_3 = \frac{5}{2}}$$

$$N \rightarrow 3\nu_1 = 2\nu_4$$

$$\boxed{\nu_4 = \frac{3}{2}}$$

$$O \rightarrow 9\nu_1 = 2\nu_2 + \nu_3 + 2\nu_5$$

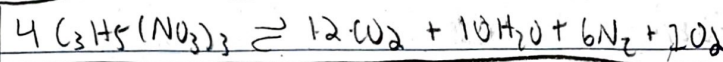
$$9 = 6 + \frac{5}{2} + 2\nu_5 \quad \boxed{\nu_5 = \frac{11}{4}}$$

$$C \rightarrow \nu_2 = 3$$

$$H \rightarrow 2\nu_3 = 5$$

$$N \rightarrow 2\nu_4 = 3$$

$$O \rightarrow 2\nu_2 + \nu_3 + 2\nu_5 = 9$$



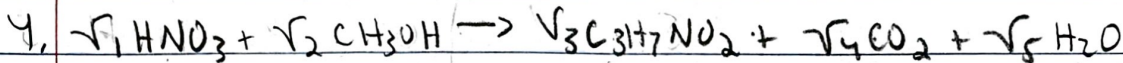
$$12C \rightarrow 12C$$

$$20H \rightarrow 20H$$

$$12N \rightarrow 12N \quad \checkmark$$

$$36O \rightarrow 36O$$

✓



$$C \rightarrow 1\nu_2 = 3\nu_3 + 1\nu_4$$

$$H \rightarrow 1\nu_1 + 4\nu_2 = 7\nu_3 + 2\nu_5$$

$$N \rightarrow 1\nu_1 = 1\nu_3$$

$$O \rightarrow 3\nu_1 + 1\nu_2 = 2\nu_3 + 2\nu_4 + 1\nu_5$$

$$\text{Set } \nu_1 = 1, \nu_3 = 1$$

$$C \rightarrow -1\nu_2 + 1\nu_4 = -3$$

$$H \rightarrow -4\nu_2 + 2\nu_5 = -6$$

$$O \rightarrow -1\nu_2 + 2\nu_4 + \nu_5 = 1$$

$$\boxed{\nu_3 = 1}$$

$$C \rightarrow -1\nu_2 - 1\nu_2 + 3\nu_3 + 1\nu_4 = 0$$

$$H \rightarrow -4\nu_2 + 7\nu_3 + 2\nu_5 = 1$$

$$N \rightarrow 1\nu_3 = 1$$

$$O \rightarrow -1\nu_2 + 2\nu_3 + 2\nu_4 + 1\nu_5 = 3$$

$$O - 2H$$

$$-1\nu_2 + 2\nu_4 + \nu_5 = 1$$

$$-2\nu_2 + 2\nu_4 = -6$$

$$\nu_2 + \nu_5 = 7$$

$$(\nu_2 + \nu_5 = 7) \times 2$$

$$-4\nu_2 + 2\nu_5 = -6$$

$$2\nu_2 + 2\nu_5 = 14$$

$$-4\nu_2 + 2\nu_5 = -6$$

$$6\nu_2 = 20$$

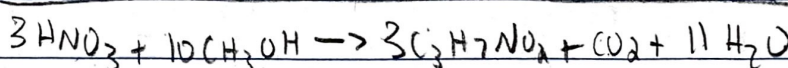
$$\boxed{\nu_2 = \frac{10}{3}}$$

$$-\frac{10}{3} + \nu_4 = -3$$

$$\boxed{\nu_4 = \frac{1}{3}}$$

$$-\frac{10}{3} + \frac{2}{3} + \nu_5 = 1$$

$$\boxed{\nu_5 = \frac{11}{3}}$$



$$54 \text{ mg/L} \times 10 = 540 \text{ mg } HNO_3 = .54 \text{ g } HNO_3$$

$$10 \text{ mg/L} \times 10 = 100 \text{ mg} = .1 \text{ g}$$

$$MM HNO_3 = 63.018 \text{ g}$$

$$.54 \text{ g} \div .1 = .44 \text{ g } HNO_3$$

$$\frac{1}{63.018} \times \frac{10 \text{ mol } CH_3OH}{3 \text{ mol } HNO_3} \times \frac{32 \text{ g } CH_3OH}{1 \text{ mol}} = 0.745 \text{ g}$$

$$MM CH_3OH = 32.0 \text{ g}$$

$$.745 \text{ g } CH_3OH$$

5. ethanol: C_2H_5OH MW = 46.075 $\frac{6.048}{46.075} \times 100\% = 13.12\%$

Methane: CH_4 MW = 16.04 g/mol $\frac{4.032}{16.04} \times 100\% = 25\%$

water: H_2O MW = 18.01 $\frac{2.016}{18.01} \times 100\% = 11.2\%$

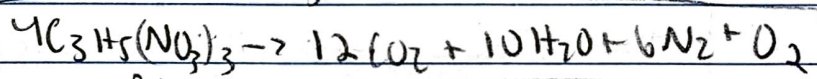
Glucose: $C_6H_{12}O_6$ MW = 180.156 $\frac{12.096}{180.156} \times 100\% = 7.75\%$

There is very little validity in the claim of withdrawing 50% wt of hydrogen in nanotubes considering nanotubes have less hydrogen weight than the above liquids.
Not likely!

3. Matrix:
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 2 & 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 3 \\ 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & | & -4 \\ 0 & 1 & 0 & 0 & | & -12 \\ 0 & 0 & 1 & 0 & | & -10 \\ 0 & 0 & 0 & 1 & | & -6 \end{bmatrix}$$

2b)



R1	R2	R3	R4	R5	R6	Error
$\frac{1}{x}$	$\frac{1}{y}$	$(x-sx)^2$	$(y-sy)^2$	$(x-sx)(y-sy)$	$\frac{B}{A^2} + \frac{1}{A}$	$(R6+R2)^2$
.4	.05	.087	.00116	.01	.0455	This changes after we use solver to minimize error
.25	.0232	.021	$5.35 \cdot 10^{-5}$	.00106	.0304	
.1	.0154	$2.62 \cdot 10^{-5}$	$3.12 \cdot 10^{-7}$	$2.86 \cdot 10^{-6}$	.0154	Use solver!
.05	.0094	.00304	$4.24 \cdot 10^{-4}$	.00036	.010	
.025	.00493	.00642	$4.42 \cdot 10^{-5}$	.000562	.008	1) Set objective to sum of Errors (n) 2) Set constraints B and A. 3) Uncheck box to make constraints non-negative
.01	.0028	.009	$6.61 \cdot 10^{-5}$	.00077	.0064	
.005	.00167	.01	$8.6 \cdot 10^{-5}$	.00093	.006	
.001	.000606	.011	$9.77 \cdot 10^{-5}$	.00103	.0055	

SSum? (n) Sum
4) Solve to get new values for A & B

A: 184.7 B: 18.5 ✓